

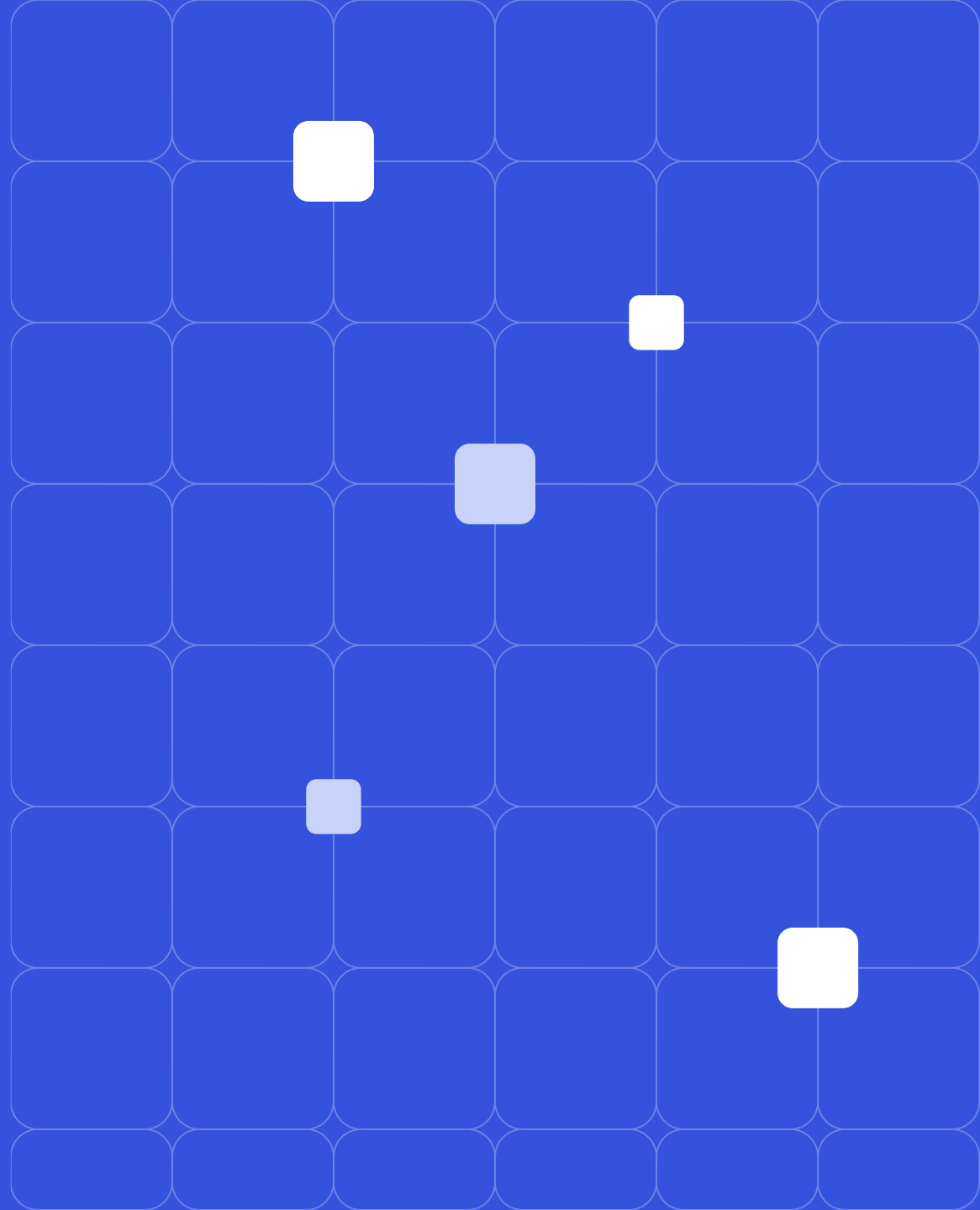
W3C × GS1 · ECOMMERCE FOR HUMANS AND AI AGENTS · ZURICH 2026

From Markup to Memory

Building an Open Web for AI Agents

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wordlift



WHERE WE ARE



GOOGLE



OPENAI



ANTHROPIC



MICROSOFT



HUAWEI



W3C

UCP · A2A · AP2 · ARD

ACP

MCP

NLWeb

TASP · A4P

WebMCP

co-authored by Microsoft and Google · adopted by OpenAI

DISCOVER · COMPARE · CONFIGURE · CHECK · AUTHORIZE · PURCHASE · RETRIEVE

What are these capabilities acting on?

What does the agent — the data consumer — actually find when it arrives?

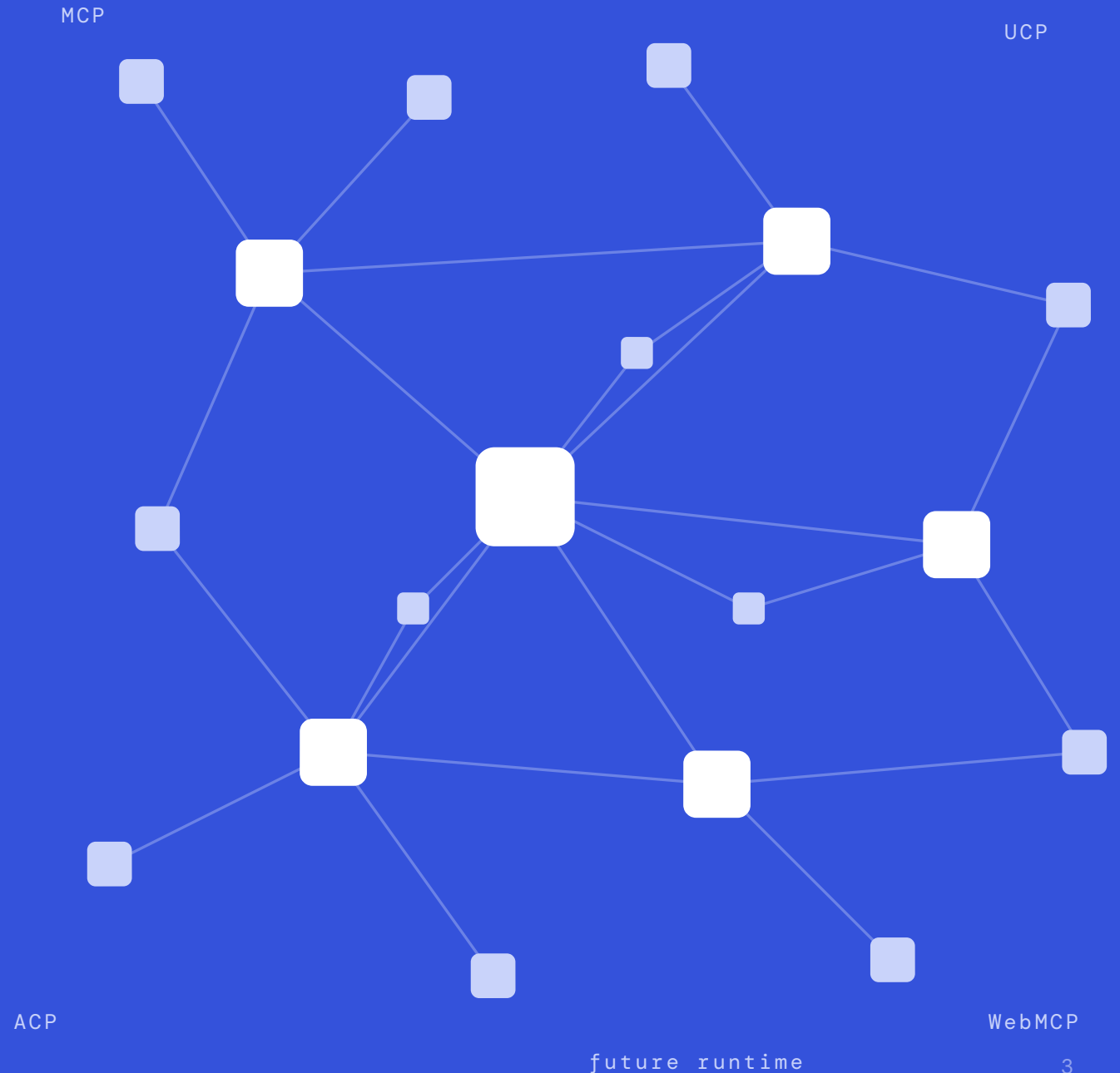
One open memory. Many agent runtimes.

The graph stays. The runtimes come and go.

THE WEB ALREADY HAS THIS LAYER

URI → HTTP → RDF → LINKS

identity · resolution · facts · relationships



So we tested it.

Schema.org reformulated Linked Data for search engines. What is the reformulation for LLMs? Same information, three exposures.

4

domains

349

questions

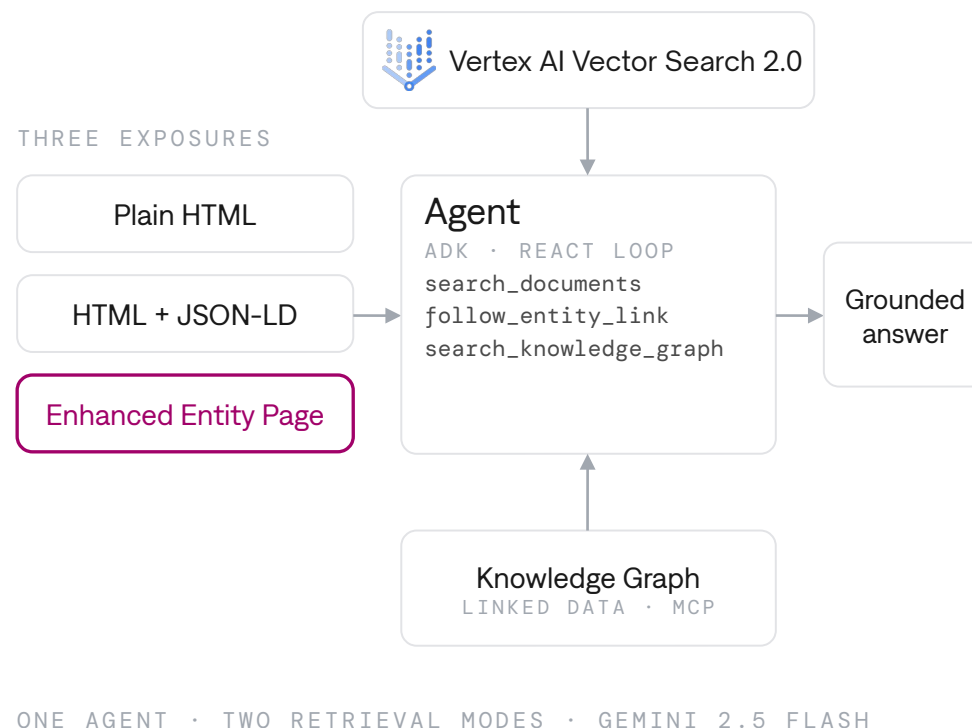
7

conditions (6 + Enhanced+)

2,443

evaluations

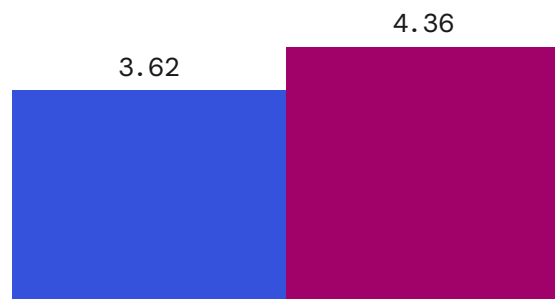
Same information in every condition. Only the exposure changed.



Markup helped. Memory changed the result.

+29.6%

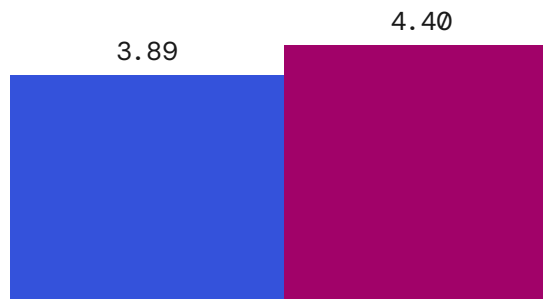
ENHANCED PAGE × STANDARD RAG · VS PLAIN HTML, STANDARD RAG



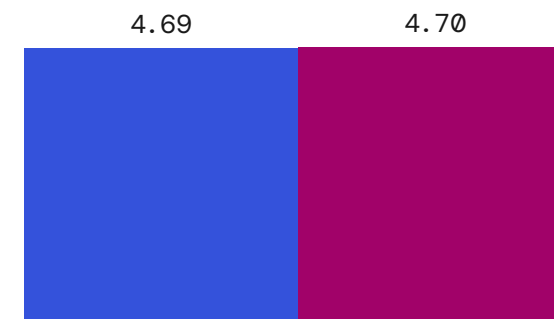
Plain HTML

+29.8%

ENHANCED PAGE × AGENTIC RAG · VS THE SAME BASELINE



HTML + JSON-LD



Enhanced Entity Page

■ Standard RAG ■ Agentic RAG

JSON-LD alone moved the score +0.27. The navigable entity page moved it +1.07. The structure had to leave the document.

THE SURPRISING RESULT

4.69 $\approx$ 4.70

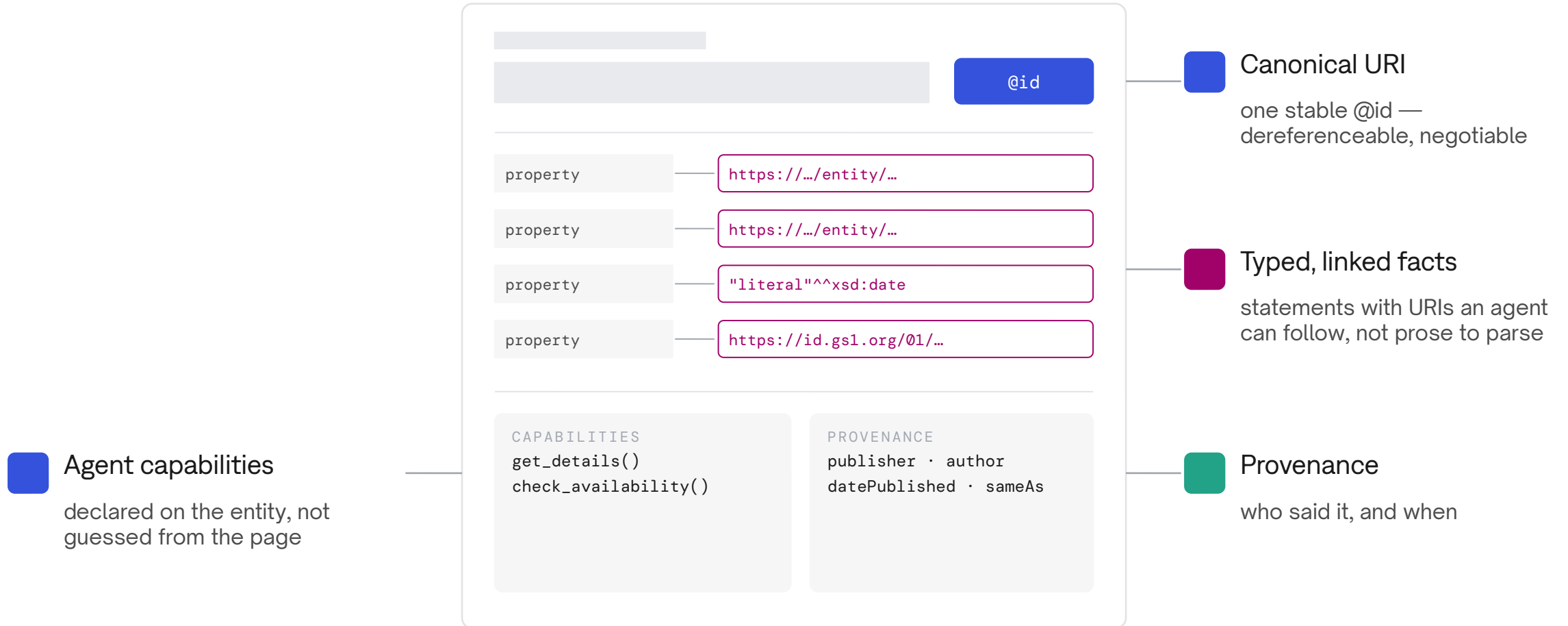
ENHANCED ENTITY PAGE × STANDARD RAG

ENHANCED ENTITY PAGE × AGENTIC RAG

Better surfaces can reduce the need for smarter retrieval.

Agents often compensate for weak information architecture. Do not push every information problem into the inference layer.

The anatomy of open memory.



And this is what it actually looks like.

Thing > Article > CreativeWork

What is a Knowledge Graph? A comprehensive Guide

Type: `schema:Article`, `schema:CreativeWork`, `schema:Thing` | [Canonical URI](#)

Description

Knowledge Graphs are a way to store both information and its meaning. It's not known when exactly the first enterprise graph was developed, though the development of knowledge graphs by Google and Facebook in organizing their content and entities.

Related Entities

Explore the knowledge graph by following these links. Each URI supports content negotiation — append `.json` for JSON-LD, `.ttl` for Turtle, or `.html` for a human-readable view.

about:	http://data.wordlift.io/wl0216/entity/google_knowledge_graph
about:	http://data.wordlift.io/wl0216/entity/artificial_intelligence
publisher:	http://data.wordlift.io/wl0216/entity/wordlift
isPartOf:	https://wordlift.io/blog/en/#website
publisher:	https://wordlift.io/blog/en/#organization

 Agent Instructions

Agent Instructions for What is a Knowledge Graph? A comprehensive Guide

> This page describes **What is a Knowledge Graph? A comprehensive Guide** as part of the WordLift Blog knowledge graph.

How to use this page

The entity as a first-class, dereferenceable, navigable Web resource.

GS1 already gives us open identity.



GTIN

stable identity



URI

Digital Link · Web resolution



GRAPH

GS1 Web Vocabulary · shared meaning

```
https://id.gs1.org/01/09506000134376
```

brand · offer · variant · manufacturer · certification · ingredients · availability · related products

GS1 speaks Linked Data. Identity and facts are only the beginning.

Identity alone is not understanding.



ENTITY

What exists?

Product · Organization · Service ·
Location · Article · Offer · Person



→

MEANING

What does it mean?

Terminology · Taxonomy · Synonyms ·
Conceptual distinctions · Constraints



→

ACTION

What can happen?

Retrieve · Compare · Check · Configure ·
Buy · Book · Escalate

customer ≠ beneficiary ≠ policy holder

frame ≠ lens ≠ variant

If the agent does not understand the organization's own terms, the existence of a tool does not guarantee the right tool is invoked.

Visibility should compile into action.

VISIBLE → UNDERSTOOD → ACTIONABLE

Entity graph

canonical ID · facts · relations · provenance

Lexical graph

terminology · taxonomy · synonyms ·
distinctions

Action graph

capability · policy · business boundary

discovered through ARD · exposed through

WebMCP · UCP · MCP

owned · partner handoff · informational only · not applicable

Being cited is not the endpoint.

The entity has to expose the next valid action.

Semantics do not automatically become capabilities.

SCHEMA.ORG · LINKED DATA

WHAT IT IS

LodgingBusiness

offers

amenityFeature

geo

WHAT IS THIS?

describes the world

ACTION GRAPH · TERMS OF ACTION

WHAT YOU CAN DO

WHO BINDS THE ENTITY TO THE RIGHT ACTIONS?

check_availability →  owned

get_details →  informational

book →  partner handoff

declared by the business, in its own terms - not derived from the schema type

will do · will describe · hands off

DEMONSTRATOR beta.audit.wordlift.io

WEBMCP + CHATGPT DEFINE THE BOUNDARIES

WEBMCP

HOW TO DO IT

get_details()

check_availability()

book() · not exposed

WHAT CAN I DO?

exposes the valid actions to the runtime

Linked Data says what it is. WebMCP says what tools exist. The missing layer binds the two.

A website can now tell the agent what it can do.
The question is where that list comes from.

ENTITY

Accommodation

schema:LodgingBusiness · sameAs · geo · amenityFeature · offers

ACTION GRAPH · FROM THE TERMS OF ACTION

get_details	→	 informational
check_availability	→	 owned
book	→	 partner handoff

WEBMCP · EXPOSED ON THE PAGE

```
navigator.modelContext.registerTool({
  name: "check_availability",
  description: "Rooms for dates and guests",
  inputSchema: { checkIn, checkOut, guests }
})

registerTool({ name: "get_details", ... })

// book(): not exposed – partner handoff

// build your own Terms of Action:
// beta.audit.wordlift.io
// github.com/wordlift/ai-audit-webmcp
// WebMCP + ChatGPT: define your boundaries in
conversation
```

The entity graph grounds the tools. The Terms of Action determine which capabilities are valid and how they are exposed.

Not only a commerce architecture.



Product

ENTITY

shade · frame · lens · store

ACTIONS

find variant · compare · locate retailer ·
check availability · buy

mostly owned



Publishing

ENTITY

article · person · event · topic · archive

ACTIONS

retrieve · follow evidence · explore
related coverage · license context

retrieval and rights



Insurance

ENTITY

product · coverage · risk · advisor · claim

ACTIONS

understand · compare · find advisor ·
request quote · handoff

not autonomous

Same open memory. Radically different action surfaces.

From public facts to licensed context.

FREE DISCOVERY LAYER

1 · DISCOVER

Enhanced Entity Page

Agent reads the public entity page: summary · facts · statements · connected entities · provenance.

FREE PUBLIC FACT SET

2 · DECIDE

Terms of Action

Are public facts enough?
Public → discovery, grounding, attribution.
Premium → full context, licensed excerpts, bundles.

FREE VS PREMIUM RULE

PAID ACCESS LAYER

3 · REQUEST OFFER

Access offer

GET /access-offer ?
entity_id={url}
&access_type={type}
Returns schema:Offer · SKU · price · license.

MACHINE-READABLE OFFER

4 · PAY

Checkout · AP2 / x402

Cart → session → AP2
mandate → payment → order ID. Intent proven, not assumed.

AUTHORIZED PAID ACCESS

5 · RETRIEVE

Licensed package

GET /access/{order_id}
Full context · related coverage · provenance · citation · license.

LICENSED ANSWER PACKAGE

Public facts create demand. Terms of Action price it. The storefront converts it into licensed context.

Do not rebuild the Web inside every agent protocol.

HTML and Linked Data remain the canonical memory layer — ours to keep evolving.
WebMCP is the open action layer taking shape at W3C.

One open memory. Many agent runtimes.

DISCOVERY

How does an agent find the canonical entity?

TRAVERSAL

How does it move from entity to related knowledge and capabilities?

BINDING

How do actions attach to entities without coupling memory to one runtime?